AI-ENABLED HR EVALUATION SYSTEM

(FROM RESUME UPLOAD TO TRANSPARENT HIRING DECISION)

Team : **Code Pipers**

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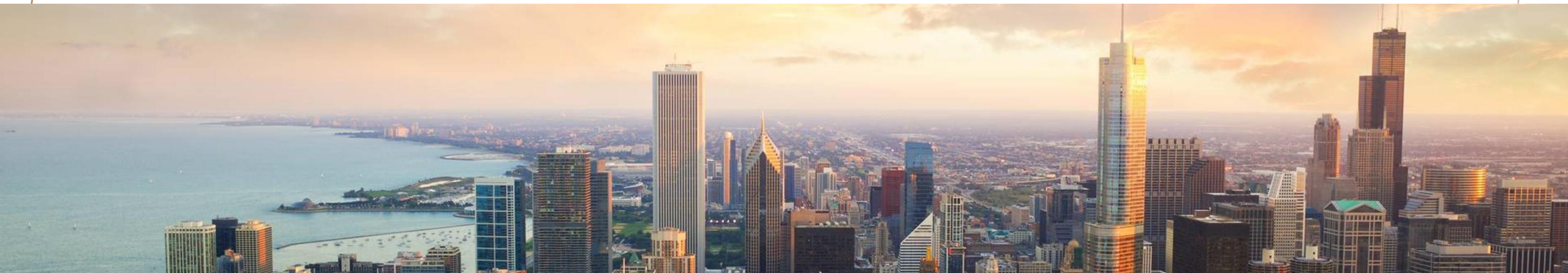
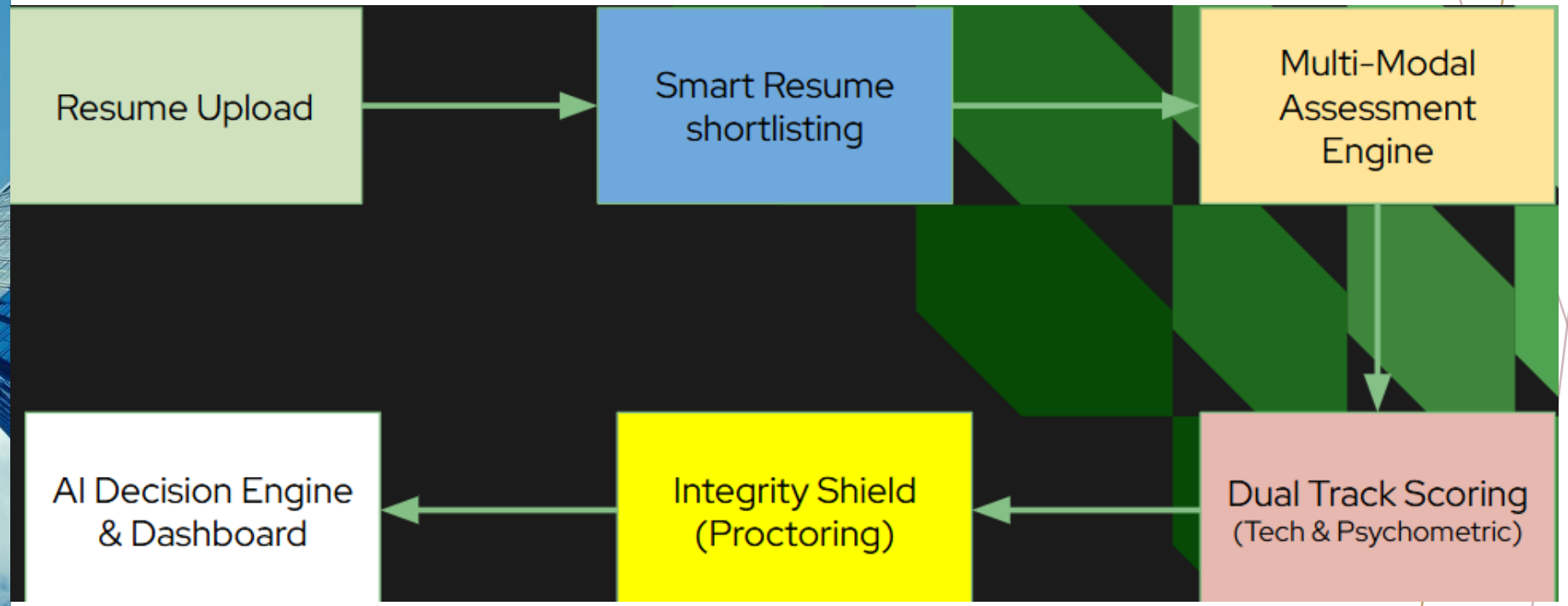
First Year – B. Tech (AI/ML) – { SRMIST }



PROBLEM & MOTIVATION

- Recruiters manually screen hundreds of resumes and can miss strong candidates.
- Technical and soft skills are often evaluated in isolation, not as a single profile.
- Remote assessments are vulnerable to cheating and proxy test takers.
- Many AI tools give a score but don’t explain why a candidate was accepted or rejected.

Solution Overview



SMART RESUME SHORTLISTING (GATEKEEPER)

Key components

- **Parsing Logic**
  - Extracts skills, years of experience, education, and relevant keywords from PDF/Word resumes.

- **Job Description Matching**
  - Compares extracted entities against a specific JD (skills, tools, role keywords).
  - Uses scoring heuristics to compute a match score.

- **Automatic Ranking**
  - Categorizes candidates into:
    - "High Match" – Strong alignment with the JD
    - "Potential" – Partial match, worth a closer look
    - "Reject" – Low alignment with the role

Input	Process	Output Category
Resume + JD	Parse -> Sore -> Rank	High Match / Potential / Reject
Input	Process	Output Category

MULTI-MODAL ASSESSMENT ENGINE

- Scenario-Based MCQ
  - Presents realistic workplace situations.
  - Candidate chooses best response.
  - Measures judgment, ethics, communication style.

- Coding Sandbox (Functional IDE)
  - In-browser coding editor where candidates write and run code.
  - Code evaluated against hidden test cases.
  - Measures syntax, logic, efficiency, and problem-solving.

- Text-Based Responses
  - Open-ended questions to understand thought process and communication.
  - Analyzed for clarity, structure, and reasoning.

- Slider-Based Inputs
  - Sliders (e.g., 1–10) for self-assessment and preferences.
  - Used for psychometric mapping: confidence, risk-taking, collaboration, etc.

DUAL TRACK ASSESSMENT LOGIC

- Technical Assessment:
  - Inputs: coding sandbox results, technical MCQs, system design questions.
- Measures:
  - Syntax accuracy
  - Algorithmic logic
  - System design understanding
  - Domain knowledge
- Psychometric Assessment:
  - Inputs: scenario-based MCQs, slider inputs, open text responses.
- Measures:
  - Emotional intelligence
  - Resilience and stress tolerance
  - Leadership and ownership
  - Culture fit and collaboration style

Track	Inputs	What it measures
Technical	Coding IDE, MCQs, design tasks	Syntax, logic, system design, domain
Psychometric	Scenarios, sliders, text answers	EI, resilience, leadership, culture fit

INTEGRITY SHIELD (PROCTORING SYSTEM)

Components

- Visual Monitoring
  - Uses webcam to detect:
    - Multiple faces in frame.
    - No face / candidate missing.
    - Suspicious or continuous unusual movement.
- Environmental Control
  - Detects browser tab switching.
  - Monitors copy-paste behavior and critical keyboard shortcuts.
  - Flags extended inactivity or abnormal patterns.
- Audit Trail
  - Logs all suspicious events with timestamp and type.
  - Recruiter dashboard shows a timeline of these events.
  - Helps decide if a candidate's score can be trusted.

AI DECISION ENGINE

Aggregate Score

Technical score

Psychometric score

Resume-JD match

Integrity risk level

Automatic Decisioning

"Hire"

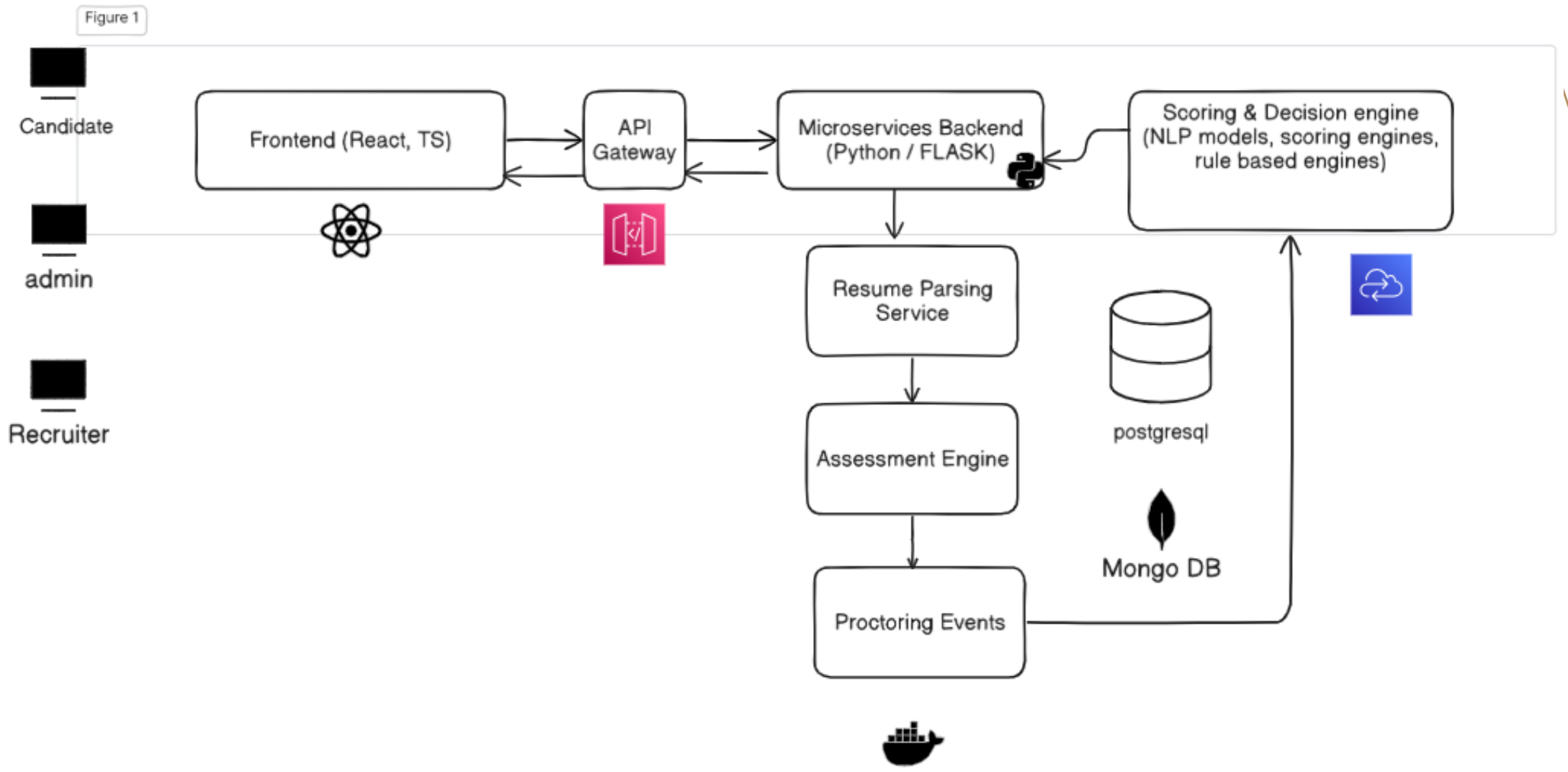
"No-Hire"

"Interview Further"

Generates explanation in plain language

"Candidate scored 90% in technical but showed low resilience and low leadership indicators in psychometric scenarios. Recommended for individual contributor role rather than leadership."

Competency Mapping:

- Visual breakdown (radar chart or bar chart style) of:
  - Strengths (e.g., Problem Solving, System Design)
  - Weaknesses (e.g., Teamwork, Stress Tolerance)
- Helps recruiters match candidate profiles to role requirements

HIGH LEVEL ARCHITECTURE